

# Saurabh Vishwakarma

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## SUMMARY

ML Engineer Python Developer — building and deploying end-to-end AI systems using PyTorch, FastAPI, and Docker. IIT Madras BS Data Science.

## PROJECT EXPERIENCE

### LeafLens — AI Image Super-Resolution Platform

Deployed · 2026

Python, PyTorch, FastAPI, Docker

[Demo](#)

- Architected and deployed an SRGAN-based super-resolution system that upscales **32×32 leaf images to 128×128** (4× factor) using a 16-block residual generator with PReLU activations,
- Implemented **8-variant Test-Time Augmentation (TTA)** — 4 rotations + horizontal flip with reverse-transform averaging — reducing prediction artifacts and improving output consistency at inference time
- Built a production **FastAPI inference backend** exposing a `/predict` endpoint (multipart/form-data) containerized end-to-end with Docker for reproducible cloud deployment on Render

### ATS Intelligence Engine

2026

Python, FastAPI, Sentence Transformers, PyMuPDF, Docker

[Source Code](#)

- Built an AI-powered resume screening system that outputs a **blended ATS score** by combining cosine similarity from Sentence Transformer embeddings with rule-based skill overlap; served via FastAPI behind an **Nginx reverse proxy**
- Designed a **dual-input REST API** supporting both raw text and PDF uploads via two endpoints (`/predict/ats` and `/predict/ats/upload`), with PyMuPDF handling unstructured PDF extraction across varied resume formats
- deployed with **Docker Compose** orchestrating FastAPI backend
- Integrated **HuggingFace Hub** for model serving — demonstrating awareness of model lifecycle:

### Language Modeling using GRU

2025

Python, PyTorch

[Source Code](#)

- Implemented a **complete GRU architecture from first principles** using raw PyTorch tensors — manually coding update gate, reset gate, candidate hidden state, and output layer — without using `nn.GRU`, to deeply understand sequential learning mechanics
- Trained on real **Amazon review text data**: built full preprocessing pipeline (tokenization → vocabulary construction → sliding window input-target pairs) and a trainable **embedding matrix** (vocab\_size × embedding\_dim) learned via backpropagation

### House Price Prediction Platform

Deployed · 2025

Python, Pandas, NumPy, Scikit-learn, Flask

[Demo](#)

- Delivered a **full regression ML pipeline** from raw data to deployed web app —
- Deployed the trained model as a **Flask REST endpoint** serving real-time predictions, demonstrating end-to-end ownership from data exploration to production-style API serving

## EDUCATION

### Chandigarh University

2023 – 2026

BCA — CGPA: 8.03

### IIT Madras

2023 – 2027

BS in Data Science — CGPA: 7.2

## TECHNICAL SKILLS

**Languages & Tools** Python, C++, SQL, Git/GitHub, Linux, Docker

**ML/DL Frameworks** PyTorch, TensorFlow, Scikit-learn, Sentence Transformers, Hugging Face Transformers, TorchVision

**Deep Learning & AI** CNNs, GANs, Computer Vision, NLP, LLMs, Transfer Learning, Image Super-Resolution, Sequence Modeling

**MLOps & Deployment** FastAPI, Flask, Docker Compose, Nginx, REST APIs, Model Serving, Render, Uvicorn

**Data Science** Pandas, NumPy, Matplotlib, Seaborn, OpenCV

**Databases & Cloud** PostgreSQL, MySQL, AWS (Basics)